

NEUROX HACKATHON

SYNAPSE

The Emotion-Aware Workspace



Team Details

Team: [The Thunder]

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01. The Problem & The Gap

!! Silent Burnout

70% of developers experience burnout.

We push through cognitive fatigue without realizing our brain is tired. This leads to high bug rates, poor architecture decisions, and severe mental exhaustion.

→ **Target:** Remote teams & students.

× Current Solutions Fail

× Pomodoro Timers

Too rigid. Interrupts flow states.

× Website Blockers

Punitive. Doesn't lower actual stress.

△ **The Gap:** We measure *Time*. We need to measure *Human Capacity*.

| 02. The Proposed Solution

An OS that actively cures burnout in real-time.

Synapse uses zero-click tracking to trigger an automatic Sensory Reset.

01.

Detection

Reads facial tension & erratic typing to map Cognitive Load.

02.

Intervention

Fades in acoustic audio & dims screen to warm tones.

03.

Empathetic OS

Pushes team health metrics to Notion Dashboard.

03. Architecture & Tech Stack

●●● System Architecture Flow

[Input] Webcam + Keyboard → [Process] Local AI Engine → [Output] UI +
Notion OS

</> Core Technologies

- **Backend:** Python, OpenCV
- **Frontend:** React, WebSockets
- **Integration:** Notion API

✓ Privacy First

Processing video locally via Python ensures **zero privacy breaches** (no video leaves the device).

WebSockets guarantee zero-latency mood shifts.

04. Key Features & Innovation

●●● Feature: The “Sensory Reset” Algorithm

Instead of just warning you, Synapse alters the environment. It cross-fades ambient focus music to artificially lower heart rates during stress spikes.

●●● Feature: Live “Team Vibe Matrix” (Notion OS)

If 70% of a hackathon team hits the “Red Zone,” the Notion dashboard automatically triggers a mandatory team-wide screen break.

●●● Feature: Privacy-First Architecture

Emotion AI can be invasive. We process locally and only send an anonymized integer (e.g., *Stress: 8*) to the Notion database.

| 05. Impact & Feasibility

○ Real-World Impact

- **Developers:** Reduces errors caused by exhaustion.
- **Managers:** Replaces fake surveys with real metrics.
- **Remote Work:** Restores the work/rest boundary.

↑ Feasibility

- ✓ **Zero Cost:** Relies entirely on built-in webcams.
- ✓ **B2B Scalable:** Enterprise licensing for Slack/Teams.
- ✓ **Hackathon Ready:** MVP can be built in 8 hours.

06. Conclusion & Demo Concept

Local Client (Python)

```
$ python engine.py  
> Tracking face landmarks...  
> [ALERT] Tension spike detected  
> Stress Level: Critical (8/10)  
> Initiating audio reset...  
> Pinging Notion API...
```

Notion OS Dashboard

Team Vibe Matrix

Live Status Sync...

User 1: **Focused**

User 2: **Stressed**

User 3: **Stressed**

[!] **Mandatory Break Initiated**

*"We aren't just building a faster workspace.
We are building an **empathetic** workspace."*